基于 liberswan 搭建 ipsec vpn

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1 产品版本

Libreswan：libreswan-4.1-1.el7_9.x86_64.rpm

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2 背景知识

Libreswan 是目前基于（"IPsec"）和 Internet 密钥交换（"IKE"）的最广泛支持和标准化 VPN 协议的自由软件实现。本文我们将使用 Libreswan 搭建点对点的 IPSec 网络，实现 OP 和 OCI 的内网互通。

2.1 Libreswan 简介

Libreswan 是 IPsec 协议的开源实现，它基于 FreeSwan 项目，可以在 RedHat 的 Linux 发行版上使用该软件包。关于 Libreswan 的 IPsec 协议的两个部分的简要信息如下所述。

基于 IPsec 的 VPN 由 Internet 密钥交换协议和封装安全有效载荷（ESP）协议组成。

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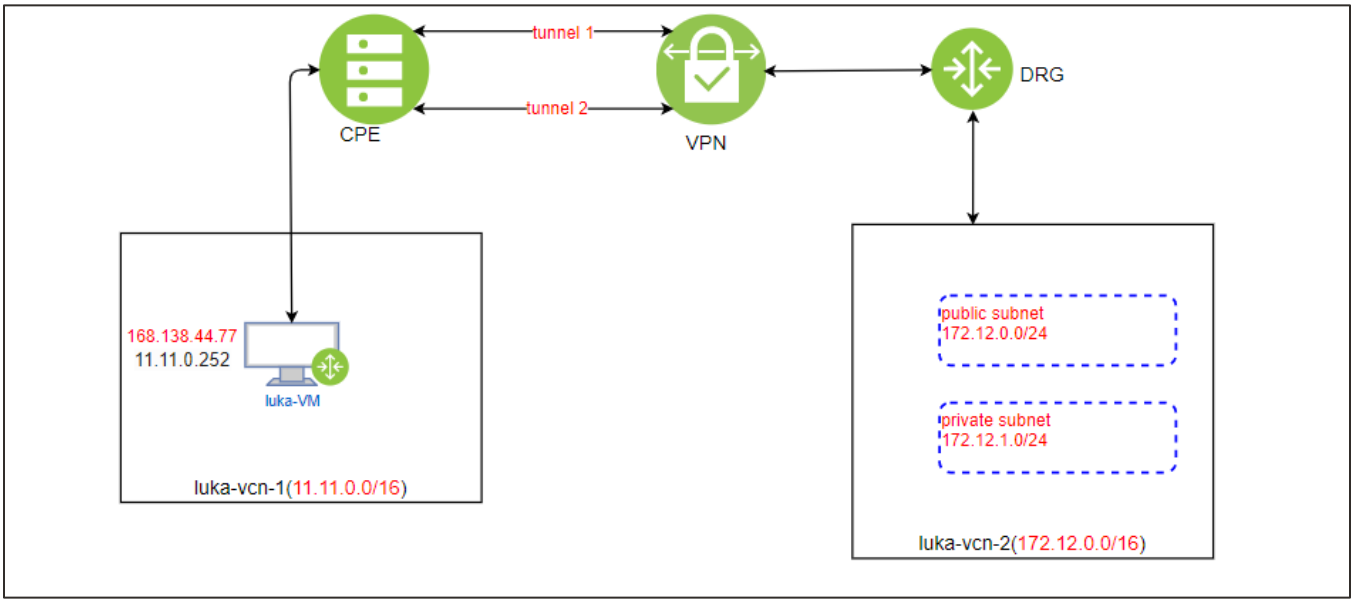
3 配置过程

测试的网络规划如下图所示

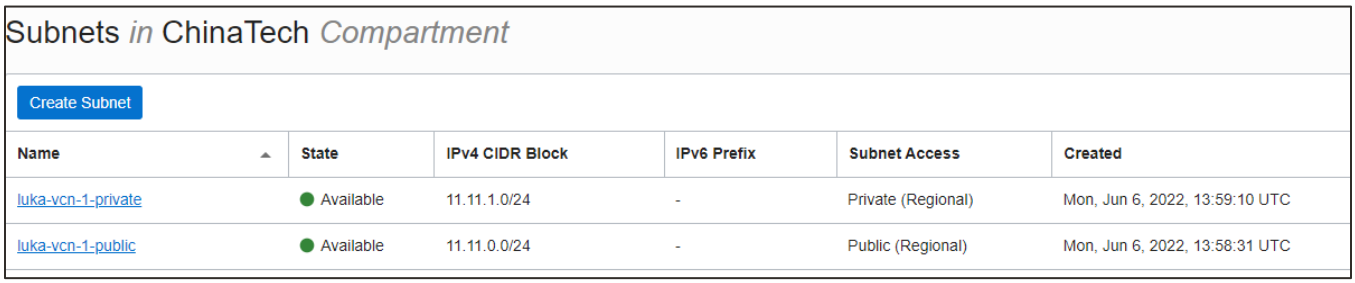


3.1 创建及配置 VCN

按照网络规划图中所示的参数，创建两条 vcn 并配置规则

3.1.1 创建 vcn，分别为 luka-vcn-1，luka-vcn-2

1. 创建 luka-vcn-1 虚拟网络及子网，创建成功如下图所示



2. 创建 luka-vcn-2 虚拟网络及子网，创建成功如下图所示

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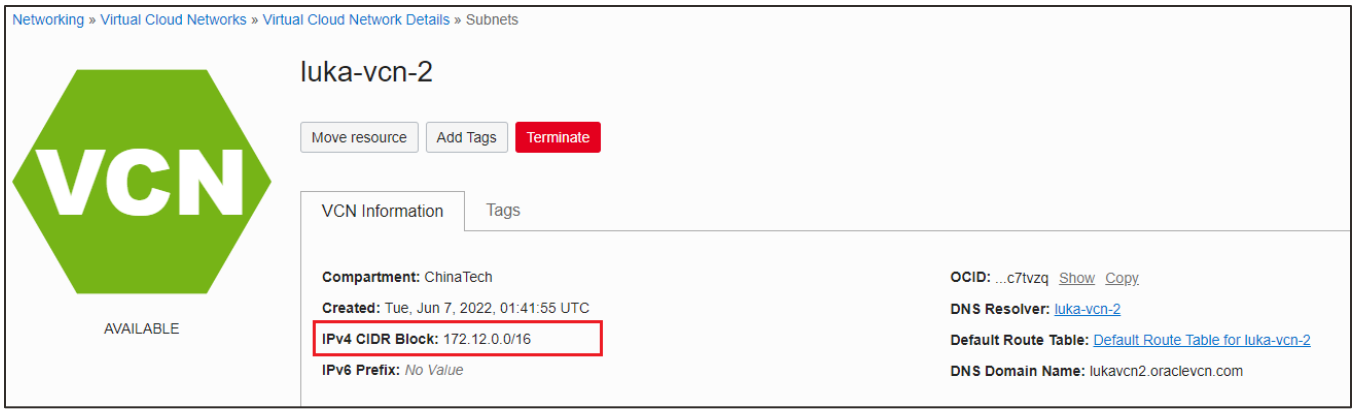
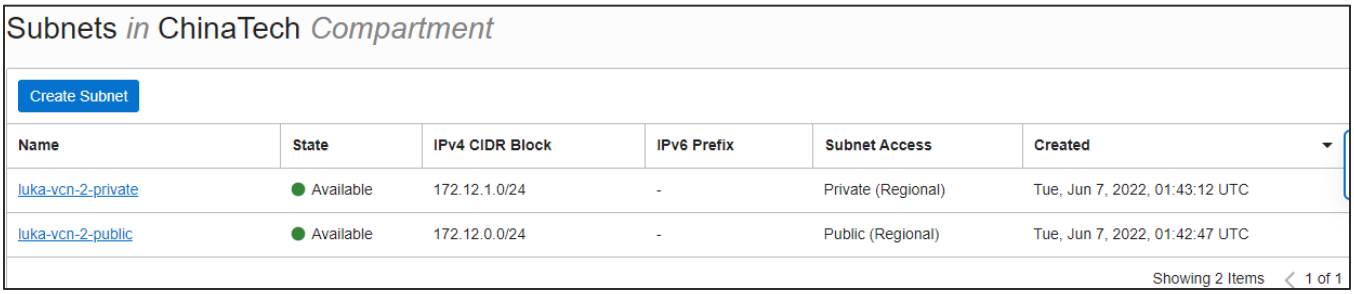
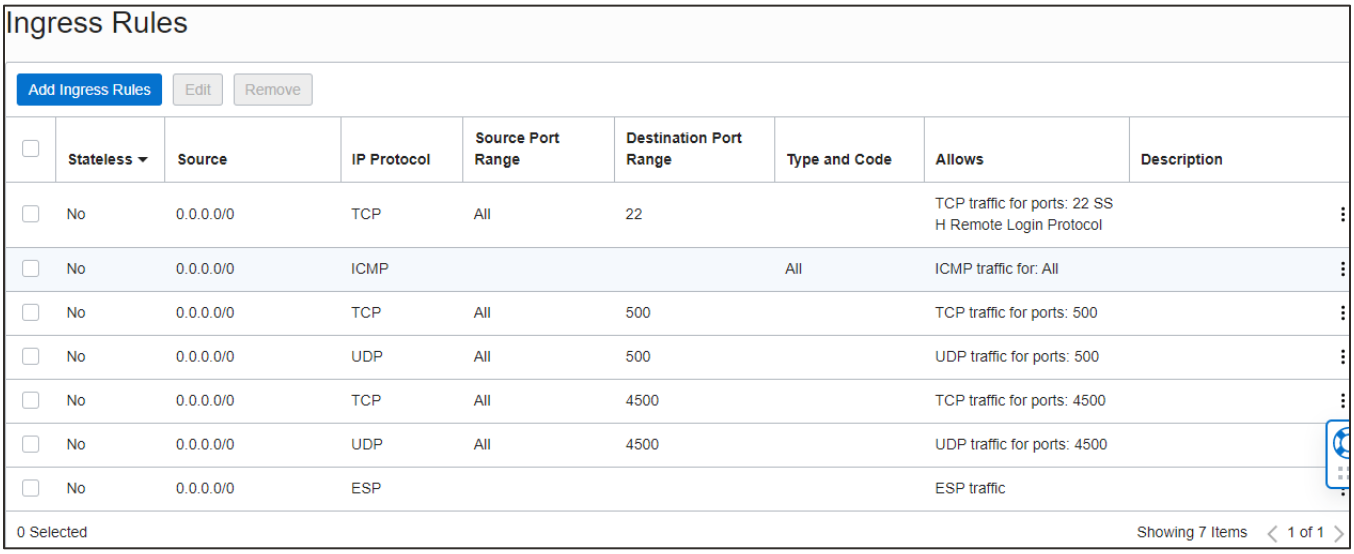
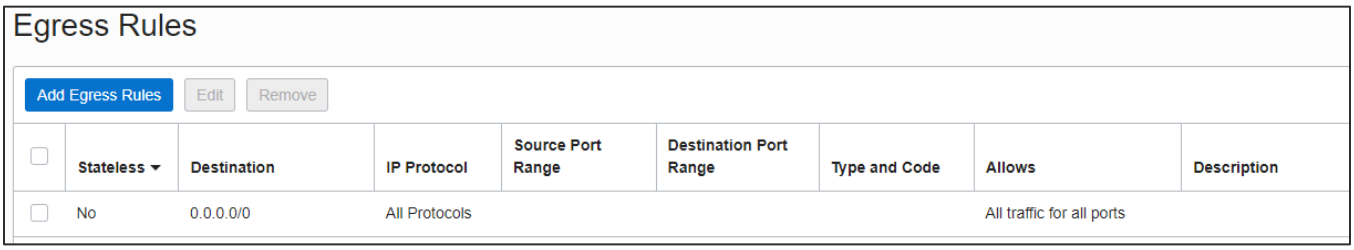




3.1.2 配置 vnc 的 security list

配置 luka-vcn-1 和 luka-vcn-2 的 security list，ingress 规则中增加对 500、4500 端口的访问权限，egress 中增加外部访问权限，具体规则见下图所示。





3.2 基于 luka-vcn-1 创建 on premise 虚拟机 luka-vm

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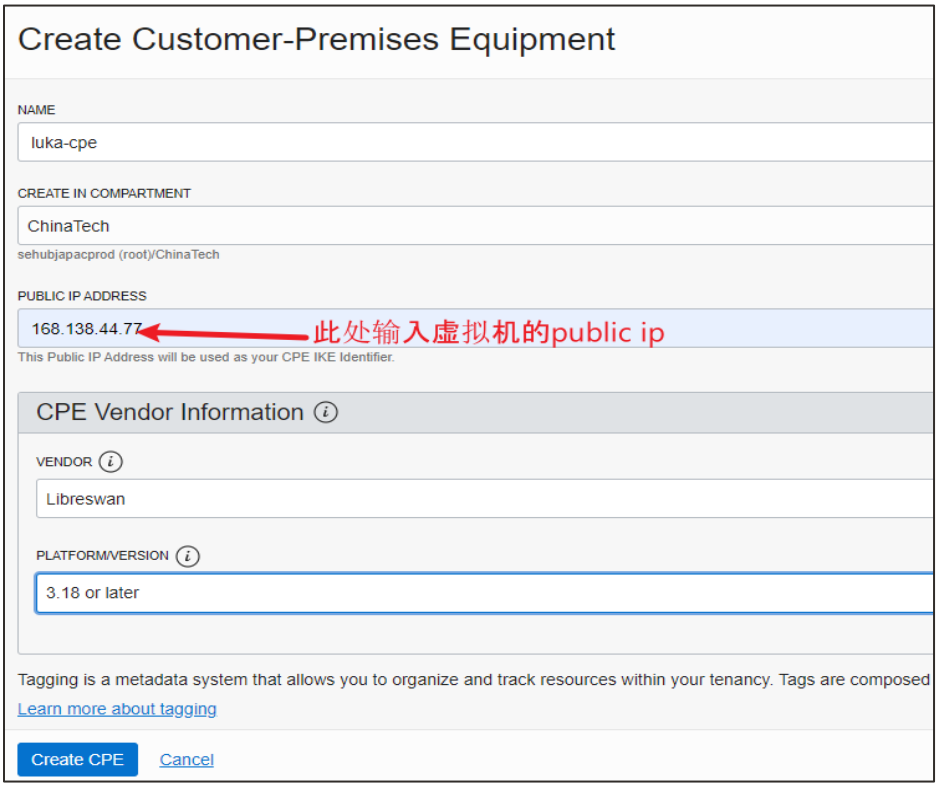
此步骤实现创建的虚拟机 luka-vm，模拟 on premise 设备

虚拟机创建过程中需要注意的参数

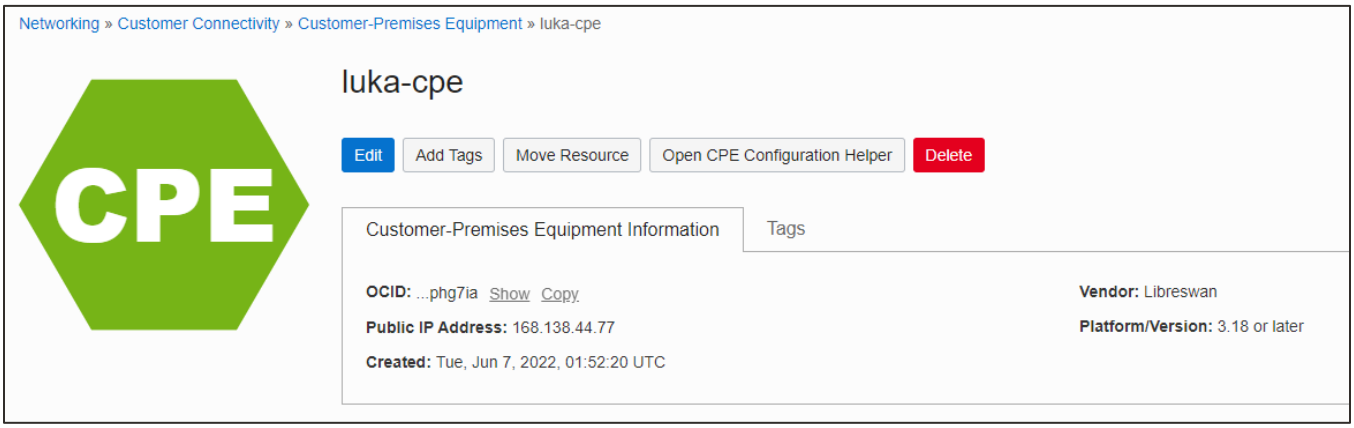
1. 使用 Oracle-Autonomous-Linux-7.9-2022.05-0 的虚拟机镜像
2. 使用 3.1 节中创建的 luka-vcn-1 网络中的 public subnet
3. 创建过程中务必执行 Upload public key files (.pub)，后续 ssh 进虚拟机执行操作使用

3.3 基于 luka-vm 创建 CPE

创建指向 Libreswan 公共 IP 地址的 customer premises equipment（CPE）对象。本例中，是基于 3.2 步创建虚拟机 luka-vm 创建 CPE，指定 luka-vm 的 public ip 为地址



创建结果如下图所示



3.4 基于 luka-vcn-2 创建 DRG

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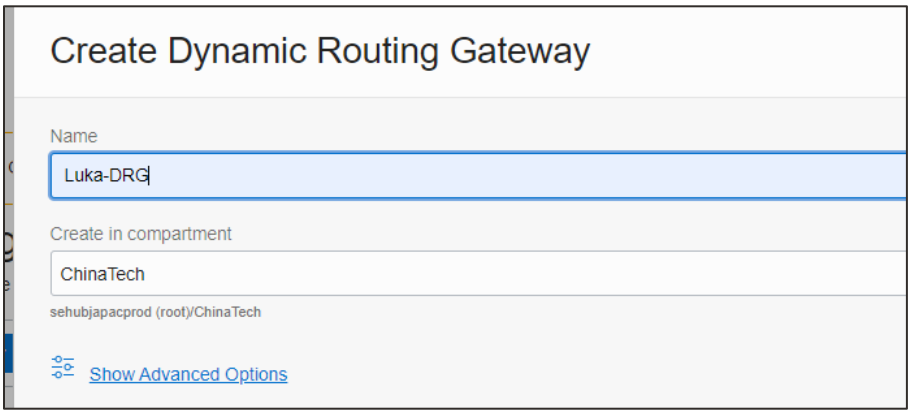
3.4.1 创建 DRG，并连接到 luka-vcn-2

1. 首先创建 DRG

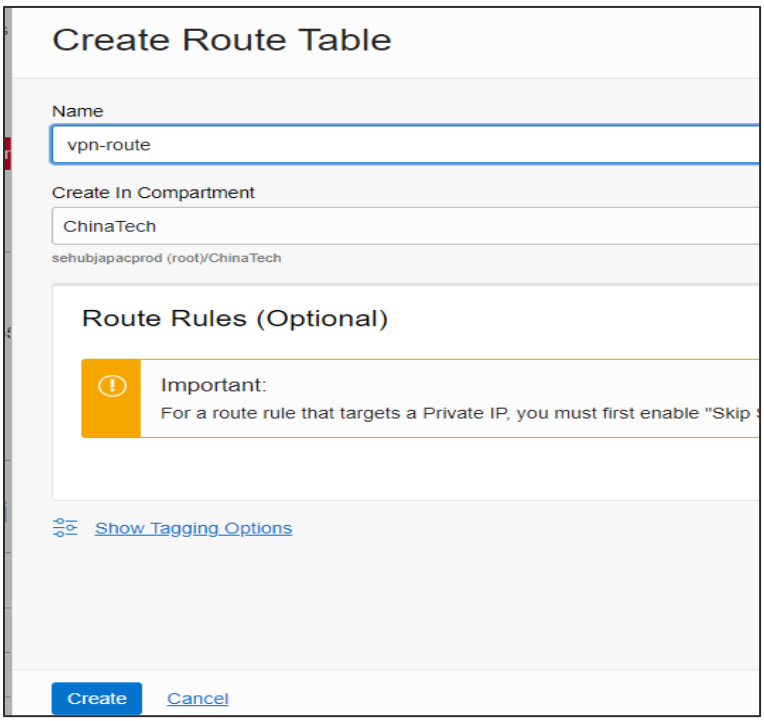


2. 待 DRG 创建成功后，选中 Virtual Cloud Networks Attachments，将 luka-vcn-2 虚拟网络和该 DRG 建立连接关系

3.4.2 基于 DRG 配置 luka-vcn-2 的 router

在 luka-vcn-2 虚拟网络中，配置到 on premise 网络的路由信息。

1. 创建 vcn 会创建一条 default 路由规则对象，但是建议此处新建一个 router，首先第一步是创建 router



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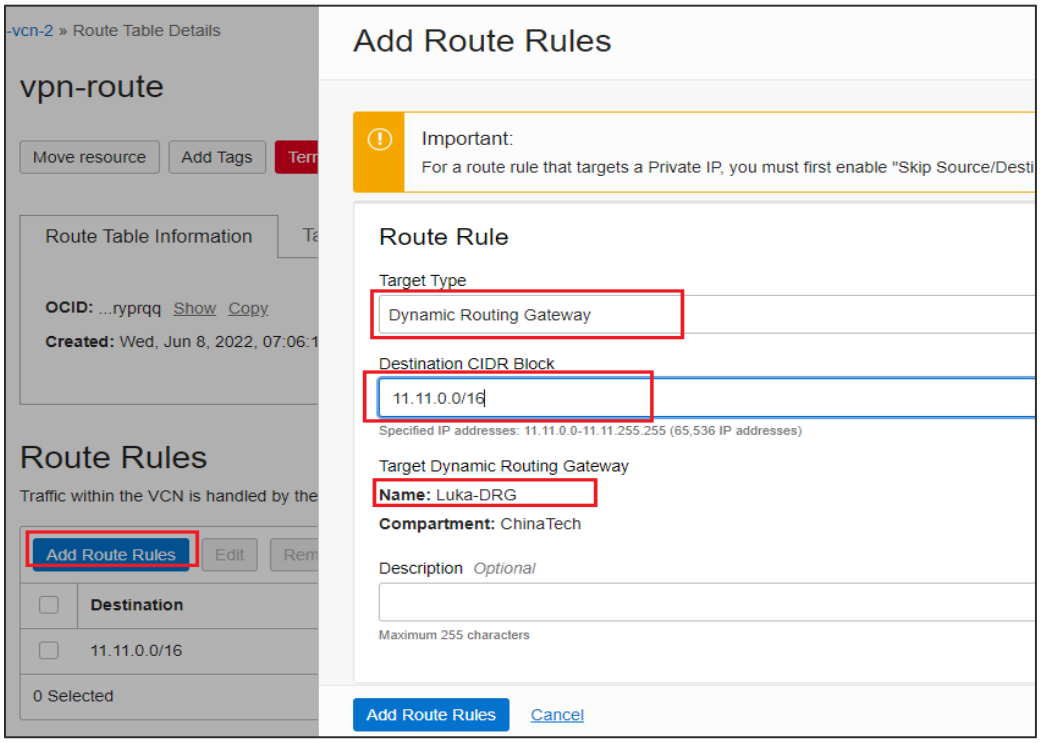
2. 在 router 创建成功后，添加一条道 on premise 网络的路由信息，并设定类型为 DRG，由于已经将 Luka-DRG 与 luka-vcn-2 建立关联，所以当选中 target type 为 Dynamic routing gateway 时，就会自动显示已经穿件好的 Luka-DRG。



3.5 创建及配置 ipsec vpn

创建 IPSec 连接，创建过程中参数具体释义

1. name 输入为 luka-ipsec
2. Customer-Premises Equipment in ChinaTech 选择在 3.3 节创建的 luka-cpe
3. Dynamic Routing Gateway Compartment in ChinaTech 选择在 3.4 节创建的 luka-DRG
4. Routes to your On-Premises Network 输入 on premise 虚拟机所在的网络 cidr，本案例为 11.11.0.0/16
5. 配置 Tunnel 1 和 Tunnel 2 参数，其中，需要记录下 Shared Secret 在后续 3.6 节使用。
   5. 1. 勾选 Provide custom shared secret，并在弹出的 Shared Secret 的输入框内设置一个秘钥，例如设置为“OCI20210408IBM”
   5. 2. IKE Version 选中 IKEv2
   5. 3. Routing Type 选择 static routing

在创建后，待 luka-ipsec 的状态显示为 available，就会在 tunnel 中显示创建两个 tunnel，状态为 down，同时，我们需要记录下这两个 tunnel 的 oracle vlan ip address 的 ip 地址在后续 3.6 节配置 libreswan 中使用。

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Tunnels *in* ChinaTech *Compartment*

Name	Lifecycle State	IPSec Status	IPV4 BGP Status	IPV6 BGP Status	Oracle VPN IP Address	Routing Type
ipsectunnel20220609065556-2	Available	Down	-	-	168.138.35.128	Static Routing
ipsectunnel20220609065556-1	Available	Down	-	-	168.138.36.15	Static Routing

Showing 2 Items

3.6 配置 vm-a 虚拟机的 liberswan

3.6.1 下载并安装 libreswan

通过如下命令，下载安装

```
sudo yum -y install https://download.libreswan.org/binaries/rhel/7/x86_64/libreswan-4.1-1.el7_9.x86_64.rpm
```

3.6.2 配置 libreswan 数据

1. 配置内核参数，修改/etc/sysctl.d/50-libreswan.conf 文件，增加如下表所示参数。其中 ens3 为 vm 的默认网卡名称，可以通过 ifconfig 查看

```
net.ipv4.ip_forward=1
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.all.send_redirects = 0
net.ipv4.conf.default.send_redirects = 0
net.ipv4.conf.ens3.send_redirects = 0
net.ipv4.conf.default.accept_redirects = 0
net.ipv4.conf.ens3.accept_redirects = 0
net.ipv4.conf.all.rp_filter = 0
net.ipv4.conf.default.rp_filter = 0
net.ipv4.conf.ens3.rp_filter = 0
```

   配置完成后，通过命令（sudo sysctl -p/etc/sysctl.d/50-libreswan.conf）触发生效

2. 通过如下表所示命令，配置防火墙规则

```
sudo iptables -I  INPUT -p tcp --dport 4500 -j ACCEPT
sudo iptables -I  INPUT -p tcp --dport 500 -j ACCEPT
sudo iptables -I  INPUT -p udp --dport 4500 -j ACCEPT
sudo iptables -I  INPUT -p udp --dport 500 -j ACCEPT
```

3. 修改 /etc/ipsec.d/oci-ipsec.conf 文件，配置 libreswan

   配置 libreswan 的配置文件中 tunnel 的具体参数，和 oci 的 ipsec vpn 建立关系，其中重要参数释义如下，其余参数默认即可。

   1. left 为 3.2 节创建的虚拟机的 private ip，leftid 为虚拟机 luka-vm 的 public ip
   2. right 为 OCI 上 3.5 节创建的 ipsec vpn 的两条隧道的 oracle vlan ip address，具体在 3.5 节中查看获取

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3.

本测试示例配置文件如下表所示

```
conn oracle-tunnel-1
    left=11.11.0.252
    leftid=168.138.44.77
    right=168.138.35.128
    authby=secret
    leftsubnet=0.0.0.0/0
    rightsubnet=0.0.0.0/0
    auto=start
    mark=5/0xffffffff # Needs to be unique across all tunnels
    vti-interface=vti1
    vti-routing=no
    ikev2=insist # To use IKEv2, change to ikev2=insist
    ike=aes_cbc256-sha2_384;modp1536
    phase2alg=aes_gcm256;modp1536
    encapsulation=yes
    ikelifetime=28800s
    salifetime=3600s

conn oracle-tunnel-2
    left=11.11.0.252
    leftid=168.138.44.77
    right=168.138.36.15
    authby=secret
    leftsubnet=0.0.0.0/0
    rightsubnet=0.0.0.0/0
    auto=start
    mark=6/0xffffffff # Needs to be unique across all tunnels
    vti-interface=vti2
    vti-routing=no
    ikev2=insist # To use IKEv2, change to ikev2=insist
    ike=aes_cbc256-sha2_384;modp1536
    phase2alg=aes_gcm256;modp1536
    encapsulation=yes
    ikelifetime=28800s
    salifetime=3600s
```

4. 修改/etc/ipsec.d/oci-ipsec.secrets，配置秘钥

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配置 libreswan 的秘钥文件，用于和 oci 的 ipsec vpn 建立关系时认证使用，由于创建的是两条隧道，所以文件中有两条记录，对应的 OCI 上 vpn 的两个隧道，每条记录的信息从左往右以此为

1. Libreswan 的 ip，也就是虚拟机的 public ip
2. 对端 OCI 上隧道 ip，具体获取方式可以参考配置 libreswan 的配置文件中的第二条，两条隧道的 ip 不同
3. PSK 开头的秘钥，获取方式为 3.5 节中第 5.1 步配置的 shared secret

本测试中示例文件如下表所示

168.138.44.77 168.138.35.128: PSK "OCI20210408IBM"
168.138.44.77 168.138.36.15: PSK "OCI20210408IBM"

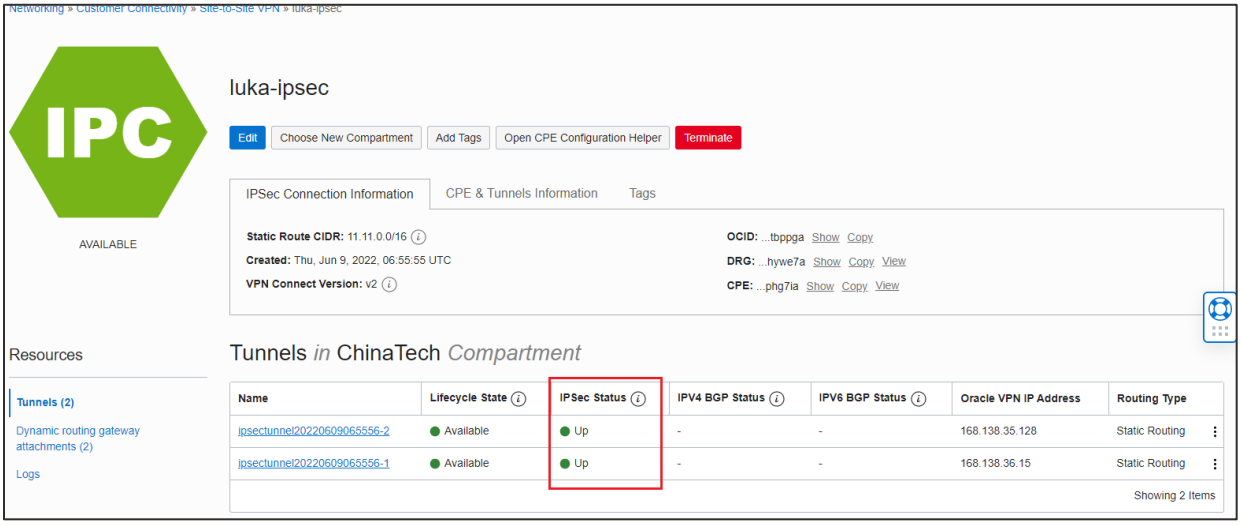
5. 配置完成后，通过命令 sudo service ipsec restart 触发 libreswan 服务重启，配置生效

```
[opc@luka-vm ~]$ sudo service ipsec restart
Redirecting to /bin/systemctl restart ipsec.service
```

3.7 验证

待完成所有配置后，可以查看 vpn 的状态，通过在 OCI 和 libreswan 进行验证

3.7.1 OCI 页面查看创建的 vpn 隧道的 status 变成 up



3.7.2 在 libreswan 验证

在 libreswan 上通过 ip tunnel show，可以查看到如下图所示，则为隧道建立成功

```
[root@luka-vm ipsec.d]# ip tunnel show
ip_vti0: ip/ip remote any local any ttl inherit nopmtudisc key 0
vti2: ip/ip remote 168.138.39.1 local 11.11.0.252 ttl inherit key 6
vti1: ip/ip remote 168.138.39.62 local 11.11.0.252 ttl inherit key 5
[root@luka-vm ipsec.d]#
```

3.7.3 虚拟机验证

1. 使用 luka-vcn-2 创建验证的虚拟机，luka-vm-2，记录 private ip

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Create instance

Name	State	Public IP	Private IP	Shape	OCPU count	Memory (GB)	Availability domain	Fault domain	Cre
luka-vm-2	Running	138.2.50.184	172.12.0.52	VM.Standard2.4	4	60	AD-1	FD-2	Tue
luka-vm	Running	168.138.44.77	11.11.0.252	VM.Standard2.4	4	60	AD-1	FD-2	Mon

2. 在 luka-vm 虚拟机（libreswan 的载体）上配置到 luka-vcn-2 的 route 信息，添加路由命令如下表所示。

```
ip route add 172.12.0.0/16 nexthop dev vti1 nexthop dev vti2
```

执行后通过 ip route 查看路由是否配置成功，如下图表示正常

```
[root@luka-vm ~]# ip route
default via 11.11.0.1 dev ens3
11.11.0.0/24 dev ens3 proto kernel scope link src 11.11.0.252
169.254.0.0/16 dev ens3 proto static scope link
169.254.0.0/16 dev ens3 scope link metric 1002
172.12.0.0/16
        nexthop dev vti1 weight 1
        nexthop dev vti2 weight 1
[root@luka-vm ~]#
```

3. 通过 ping 虚拟机（luka-vm-2）的 private ip，查看 vpn 通道是否正常，如下图表示 vpn 部署成功且通道正常。

```
[root@luka-vm ~]# ping 172.12.0.52
PING 172.12.0.52 (172.12.0.52) 56(84) bytes of data.
64 bytes from 172.12.0.52: icmp_seq=1 ttl=253 time=1.16 ms
64 bytes from 172.12.0.52: icmp_seq=2 ttl=253 time=1.18 ms
64 bytes from 172.12.0.52: icmp_seq=3 ttl=253 time=0.675 ms
^C
--- 172.12.0.52 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 0.675/1.006/1.184/0.235 ms
```

3.7.4 验证

实现 luka-vcn-1 网络的虚拟机都可以访问通过 vpn 访问 luka-vcn-2 网络中的虚拟机（例如 luka-vm-2）

1. 使用 luka-vcn-1 网络创建虚拟机 luka-vm-3，和 luka-vm-1 在同一个 vcn，现需要实现访问 luka-vm-2 的 ip

Create instance

Name	State	Public IP	Private IP	Shape	OCP
luka-vm-3	Running	138.2.38.248	11.11.0.217	VM.Standard2.4	4
luka-vm-2	Running	138.2.50.184	172.12.0.52	VM.Standard2.4	4
luka-vm	Running	168.138.44.77	11.11.0.252	VM.Standard2.4	4

2. 在 luka-vcn-1 网络的路由表中，设置以 luka-vm-1 的 private ip 为下一跳的目的为 luka-vcn-2 的 cidr 的路由信息，具体配置如下图所示

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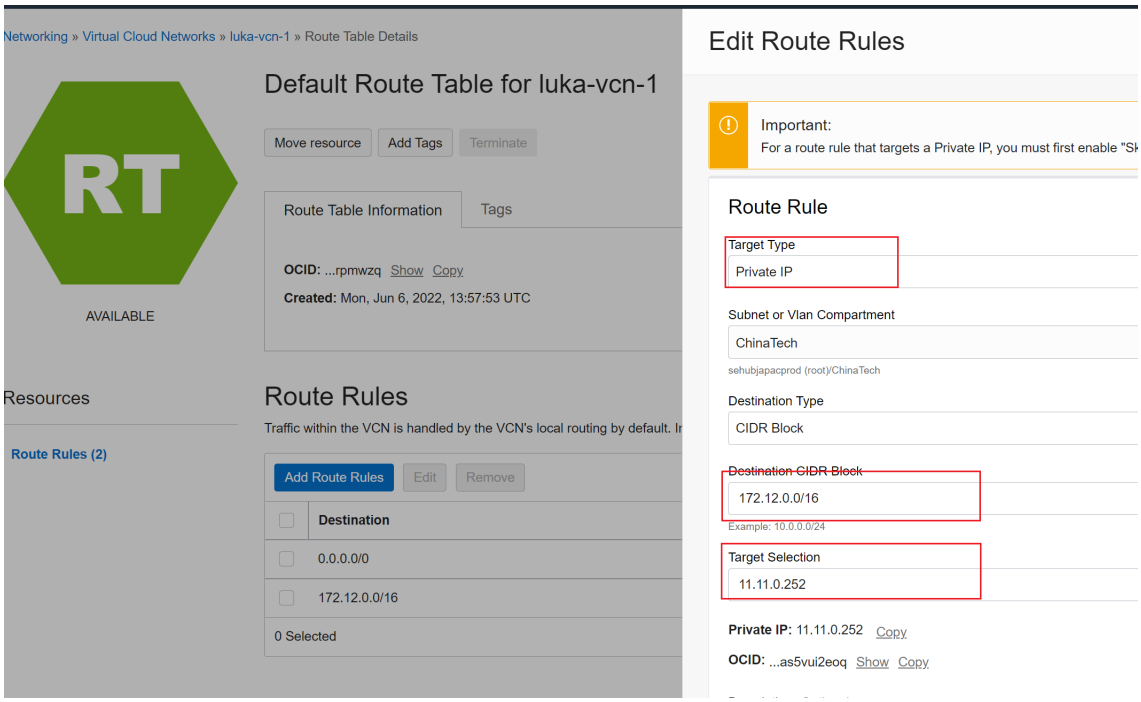

3. 在 luka-vm-1 的虚拟机内部，启动 ip 转发

通过 sysctl 可以开启 ipv4 的转发功能 (无需重启)： sysctl -w net.ipv4.ip_forward=1 或者 echo 1 > /proc/sys/net/ipv4/ip_forward 这种设置只是暂时的，它的效果会随着计算机的重启而失效。 通过在/etc/sysctl.conf 设置参数： 如果想使 IP 转发永久生效，就请修改/etc/sysctl.conf ，在这里可以增加一条 net.ipv4.ip_forward = 1 /etc/sysctl.conf: net.ipv4.ip_forward = 1 如果 ipv4 转发项已被设为 0 那么你只需要将它改为 1。

4. 通过如下命令，启动防火墙的端口转发功能。访问指定的 ip+端口时，可以将流量转发至指定其他指定的 ip+端口。端口转发操作必须启用区域的 ip 伪装功能。

```
firewall-cmd --zone=public --add-masquerade
```

5. 登录到 luka-vm-3 虚拟机，执行 ping 操作查看是否可以访问 luka-vm-2 的 private ip

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```
[root@luka-vm-3 ~]# ifconfig ens3
ens3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 9000
        inet 11.11.0.217  netmask 255.255.255.0  broadcast 11.11.0.255
        inet6 fe80::17ff:fe01:73f0  prefixlen 64  scopeid 0x20<link>
        ether 02:00:17:01:73:f0  txqueuelen 1000  (Ethernet)
        RX packets 90566  bytes 701882179 (669.3 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 87163  bytes 78198641 (74.5 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[root@luka-vm-3 ~]# ping 172.12.0.52
PING 172.12.0.52 (172.12.0.52) 56(84) bytes of data.
64 bytes from 172.12.0.52: icmp_seq=1 ttl=252 time=1.07 ms
64 bytes from 172.12.0.52: icmp_seq=2 ttl=252 time=1.03 ms
64 bytes from 172.12.0.52: icmp_seq=3 ttl=252 time=0.907 ms
^C
--- 172.12.0.52 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 0.907/1.002/1.067/0.073 ms
[root@luka-vm-3 ~]#
```

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